

# Shuangning Yang

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🌐 [github.com/VodkaVortex](https://github.com/VodkaVortex)    🌐 [Personal Website Coming Soon]

## RESEARCH INTERESTS

My research interests lie at the intersection of artificial intelligence and software security, with a focus on two interrelated questions: (1) how to strengthen the reliability, robustness, and trustworthiness of LLMs and the systems built upon them, and (2) how to leverage LLMs to automate and enhance security tasks.

## EDUCATION

- Anhui University, M.Eng. in Artificial IntelligenceExpected May 2027
  - Average Scores: 92/100    Research Focus: LLM Security & AI for Vulnerability Detection
- Hefei University of Technology, B.Eng. in Logistics ManagementSep. 2020 – Jul. 2024
  - GPA: 3.8/4.3    Track: IoT & Information Systems
  - Selected Coursework: C Programming, Java, Computer Networks, Database Systems

## PUBLICATIONS

- S. Yang, H. Yang, G. Zhao, M. Zhou, J. Tai, and X. Jia, “What is the Risk? Evaluating the Impact of Knowledge Distillation on LLM Vulnerabilities,” in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2026.
- H. Yang, J. Guo, S. Yang, G. Zhao, Q. Liu, C. Zhang, Z. Tan et al., “IoTbec: An Accurate and Efficient Recurring Vulnerability Detection Framework for Black Box IoT Devices,” in *Proc. Network and Distributed System Security Symp. (NDSS)*, 2026.
- S. Yang, H. Yang, J. Peng et al., “DetFuzz: A Semantic-Guided Fuzzing Framework via Operation Logic Inference for Black-box IoT Devices,” *Under Submission*.
- J. Tai, S. Yang, J. Wang, Y. Li, Q. Liu, and X. Jia, “Survey of Adversarial Attacks and Defenses for Large Language Models,” *Journal of Computer Research and Development*, 2025.
- Q. Liu, H. Yang, S. Yang, G. Zhao, J. Peng, J. Guo, and W. Zhang, “FirmSV: Detecting Stored Vulnerabilities in IoT Firmware using Static Taint Analysis,” in *Proc. Conf. Computer Supported Cooperative Work in Design (CSCWD)*, 2026.

## RESEARCH EXPERIENCE

- Semantic-Guided Fuzzing for Black-box IoT DevicesOct.2025 – Feb.2026  
*Lead Researcher*

- Designed a semantic-guided fuzzing pipeline that constructs operation-logic-to-parameter mappings from IoT firmware to automate valid test case generation.
- Detected **137 vulnerabilities** (122 CVE IDs assigned; **40 previously unknown**, avg. CVSS **8.54**); outperformed SOTA black-box and grey-box fuzzers by **7×**, achieving 100% precision and 94.48% recall.

- Security Risks of Knowledge Distillation in LLMsApr.2025 – Sep.2025  
*Lead Researcher*

- Designed and implemented an evaluation framework across five distillation configurations (SeqKD, PPO/DPO-based RL/RO and their combinations) with dual-judge assessment (human + GPT-4), demonstrating that joint knowledge learning increases jailbreak ASR by up to 18.38% and induces alignment failure in safety-robust models.
- Constructed **JBB-HarmLevel**, an extended jailbreak benchmark with 10 malicious query categories and fine-grained harm-level annotations, enabling category-level analysis of distillation-induced risk.
- Published as first author at **ICASSP 2026**.

- IoT Vulnerability Detection and Firmware AnalysisMay.2025 – Aug.2025  
*Research Contributor*

- Conducted manual vulnerability discovery following the IoTbec methodology on real-world IoT devices, identifying **~33 vulnerabilities** and submitting **20+ independent CVEs** (not counted in DetFuzz).
- Conducted literature survey on recurring vulnerability detection; contributed to manuscript writing and visualization for **IoTbec** (NDSS 2026, CCF-A) and **FirmSV** (CSCWD 2026).

## PROJECTS

- Auto-1day: An End-to-End LLM Agent for IoT Firmware Vulnerability Discovery2025 – Present  
*Independent Developer · Open Source*

- Context:** IoT firmware 1day vulnerability mining requires coordinating multiple expert tools (disassembler, taint analyzer, fuzzer, emulator) and involves repetitive manual work that bottlenecks security researchers, with high false-positive rates from existing static analysis tools (60–80%).
- Goal:** Designed and built an open-source multi-agent system that automates the full 1day discovery pipeline—from CVE patch analysis to PoC generation and emulator-based verification—targeting commodity IoT devices (Tenda, TP-Link, TOTOLink).

- **Approach:** Architected a three-layer system (orchestration / capability / tool) with structured agent communication; integrated IDA Pro via MCP for binary analysis, taint analysis frameworks for candidate identification, and a custom firmware emulation environment for dynamic verification; implemented LLM-based semantic filtering on taint-analysis output to suppress false positives, and template-driven PoC generation for command-injection and stack-overflow classes.
- **Outcome:** [在这里填你的实际成果，例如：reduced FP rate from X% to Y% on benchmark CVEs / discovered N previously-unreported 1day vulnerabilities across M firmware versions / open-sourced at [github.com/VodkaVortex](https://github.com/VodkaVortex) with X stars].

## SKILLS

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- **Programming:** Python, C, Java
- **Security Tools:** IDA Pro, QEMU, Binwalk
- **AI / ML:** Agent Programming, PyTorch, HuggingFace Transformers (fine-tuning, RL-based training)
- **Language:** Mandarin (Native), English (Conversational)
- **Certifications:** CISP-PTE (Certified Information Security Professional – Penetration Testing Engineer), China Information Technology Security Evaluation Center